Solution

Let x = length of the cut on each side of the little squares.

V = the volume of the folded box.

The length of the base after two cuts along each edge of size x is $12-2x$. The depth of the box after folding is x, so the volume is $V = x\times(12-2x)^2$. Note that, when $x=0$ or 6, the volume is zero and hence there cannot be a box. Therefore the problem is to maximize, $V = x\times(12-2x)^2, x\in(0,6)$.

Now, $$\frac{dV}{dx} = (12-2x)^2 - 4x(12-2x)$$

$$= (12-2x)(12-6x).$$

$\frac{dV}{dx}=0$ gives the stationary numbers $x=2,6$. Since $6\notin(0,6)$ the only stationary number is at $x=2\in(0,6)$. Further, $\frac{dV}{dx}$ changes its sign from postive to negative when passing through $x=2$. Therefore at $x=2$ the volume V is local maximum. The local maximum volume value is $V=128$ units. Hence the maximum cut can only be 2 units.

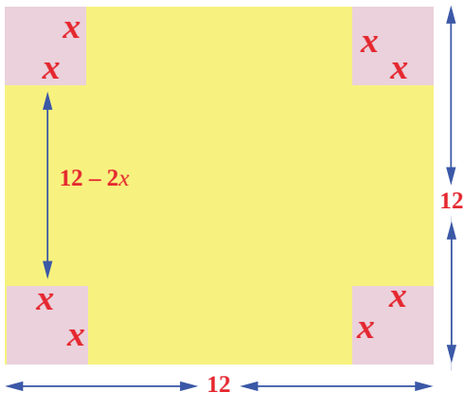


Fig.7.27

Example 7.63

Find the points on the unit circle $x^2+y^2=1$ nearest and farthest from $(1,1)$.

Solution

The distance from the point $(1,1)$ to any point (x,y) is $d=\sqrt{(x-1)^2+(y-1)^2}$. Instead of extremising d, for convenience we extremise $D=d^2=(x-1)^2+(y-1)^2$, subject to the condition $x^2+y^2=1$. Now, $\frac{dD}{dx}=2(x-1)+2(y-1)\times\frac{dy}{dx}$, where the $\frac{dy}{dx}$ will be computed by differentiating $x^2+y^2=1$ with respect to x. Therefore, we get $2x+2y\frac{dy}{dx}=0$ which gives us $\frac{dy}{dx}=-\frac{x}{y}$.

Substituting this, we get $$\frac{dD}{dx} = 2(x-1)+2(y-1)\left(-\frac{x}{y}\right)$$

$$= \frac{2[xy-y-xy+x]}{y}$$

$$\frac{dD}{dx} = 2\left[\frac{x-y}{y}\right]=0$$

$$\Rightarrow x=y$$

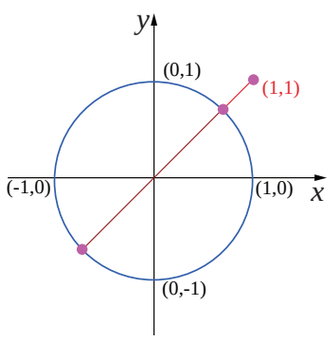


Fig.7.28

Since (x,y) lie on the circle $x^2+y^2=1$, we get $2x^2=1$ which gives $x=\pm\frac{1}{\sqrt{2}}$. Hence the points at which the extremum distance occur are, $\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right),\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$.

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